PHILIPP BAYER, PHD

COMPUTATIONAL BIOLOGY & AI

CONTACT

0435 626 108

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LinkedIn profile, Github profile

35B Archdeacon Street, Nedlands WA

SKILLS

More than ten years of research, technical, and academic leadership

Highly proficient in Python and R to build technical solutions

Highly developed communication skills

Experience in developing and deploying interpretable ML-based solutions

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EDUCATION

PhD, Applied Bioinformatics

University of Queensland

2012-2016

Novel applications of genotyping by sequencing in plant breeding

Master, IT

Bond University

2010-2012

BSc, Biology

University of Münster

2006-2009

REFERENCES

Dr Sebastian Rauschert
0481 966 400
seb.rauschert@gmail.com

Prof. Dave Edwards
0423 826 042
dave.edwards@uwa.edu.au

PROFILE

I work at the intersection of artificial intelligence, large-scale data science, and computational biology. I lead teams that build trustable, interpretable pipelines and AI tools that let government agencies and the public understand complex biological data.

I have extensive experience in reproducible computational research using Python, R, and high-performance computing clusters. I have co-published more than 180 publications that have been cited by other researchers more than 13,000 times, and have co-supervised 15 PhD students to completion.

WORK EXPERIENCE

Principal, Research Computational Biology

Minderoo, OceanOmics — 2022-current

- Mission: making eDNA-based results reliable and trustable for governments
- Developed and managed grants worth more than $ 1 million with academic partners
- Built internal quality control pipelines, AI-based summary and curation tools, led development of internal eDNA-analysis pipelines and visualisation dashboards
- Worked closely with international eDNA communities such as OBON and FAIRe

DECRA Fellow

University of Western Australia — 2021-2022

- One of 200 successful yearly applicants in Australia
- Using interpretable machine learning to identify what drives gene birth and gene death in plant genomes
- Working closely with plant breeding industry to improve breeding pipelines
- Member, Scientific Advisory Panel UWA Data Institute
- Finalist, WA Premier's Science Awards for early career researchers

Forrest Fellow

University of Western Australia — 2017-2020

- One of three first-ever Forrest Fellows chosen out of more than 400 applicants
- Independent grant to establish postdoctoral studies
- Plant pangenomics to identify highly variable gene regions in plants
- Two grants ($829,000) with Grains Research Development Corporations to develop better plant breeding methods using image-based AI and genomics

Postdoctoral researcher

University of Western Australia — 2015-2017

- Working on plant pangenomics, analysing >1000 plant genomes to identify variable genes
- Established Hacky Hour, UWA-wide data and coding community
- Became a certified Carpentries instructor to lead coding workshops for researchers